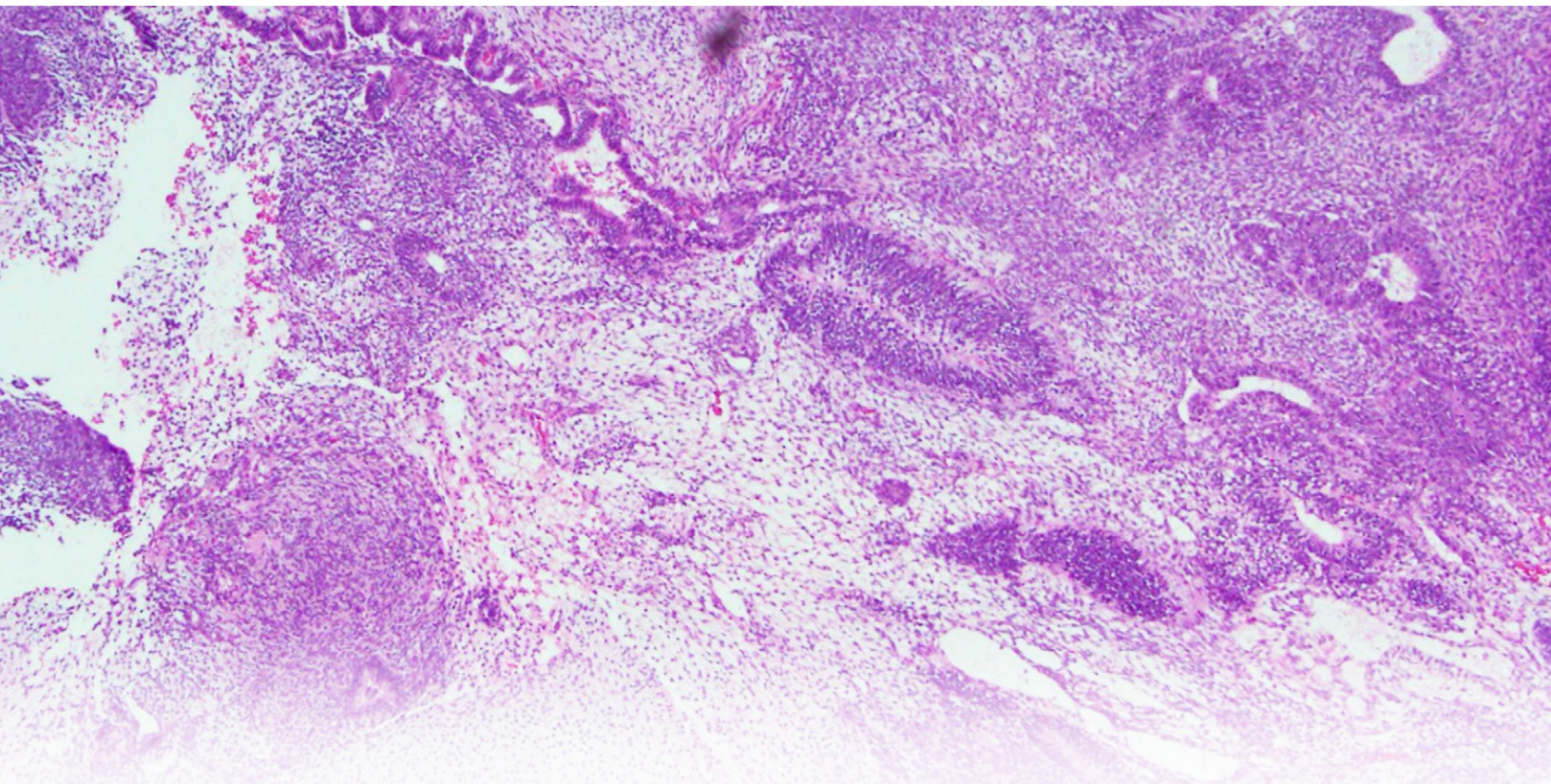


Cardiac and Pericardial Neoplasms in Children: Radiologic-Pathologic Correlation

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Primary cardiac and pericardial neoplasms are rare in the pediatric population and can include both benign and malignant lesions. Rhabdomyomas, teratomas, fibromas, and hemangiomas are the most common benign tumors. The most common primary cardiac malignancies are soft-tissue sarcomas, including undifferentiated sarcomas, rhabdomyosarcomas, and fibrosarcomas. However, metastatic lesions are more common than primary cardiac neoplasms. Children with primary cardiac and pericardial tumors may present with nonspecific cardiovascular symptoms, and their clinical presentation may mimic that of more common nonneoplastic cardiac disease. The diagnosis of cardiac tumors has recently been facilitated using noninvasive cardiac imaging. Echocardiography is generally the first-line modality for evaluation. Cardiac MRI and CT are used for tissue characterization and evaluation of tumor size, extension, and physiologic effect. The varied imaging appearances of primary cardiac neoplasms can be explained by their underlying abnormality. Treatment of these lesions varies from conservative management, with spontaneous regression of some lesions such as rhabdomyomas, to surgical resection, particularly in patients with associated heart failure. With adequate imaging techniques and knowledge of the pathologic basis of the neoplasm, it is often possible to differentiate benign from malignant tumors, which can greatly affect adequate and timely treatment.

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Introduction

Primary cardiac and pericardial neoplasms are rare; however, they are a diagnostic challenge for clinicians because they can be asymptomatic or mimic other more common cardiovascular

abnormalities such as arrhythmia, congestive heart failure, valvular insufficiency, or murmurs (1). The clinical presentation and prognosis of patients with these lesions vary depending on lesion location and histologic type (Fig 1). The prevalence of



Quiz questions for this article are available through the Online Learning Center.

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Abbreviations: FDG = fluorine 18 fluorodeoxyglucose, H-E = hematoxylin-eosin, IMT = inflammatory myofibroblastic tumor, LGE = late gadolinium enhancement, SSFP = steady-state free precession

TEACHING POINTS

- The majority of primary cardiac neoplasms in children are rhabdomyomas (60%), which are congenital hamartomas. Up to 80% of cases are associated with tuberous sclerosis, a genetic disease characterized by the presence of multiple benign hamartomas in several organs.
- The majority of rhabdomyomas regress spontaneously in early childhood. Thus, first-line management is often conservative, and surgical resection is reserved for patients with masses that result in hemodynamic disturbances.
- Cardiac fibromas are the second most common type of benign cardiac tumors in children after rhabdomyomas. They are associated with Gorlin syndrome in up to 3% of cases.
- Sarcomas can manifest anywhere in the heart, but some lesions have a higher preference for the left atrium, including undifferentiated sarcomas. Rhabdomyosarcoma is the most common primary cardiac malignancy of childhood.
- Mediastinal masses such as teratomas and thymolipomas are masses that should be considered when evaluating pericardial lipomas and lipoblastomas. Mediastinal teratomas have fat but also have fluid-attenuation cystic components and calcification, which are features that can help to differentiate these masses from lipomas and lipoblastomas.

primary cardiac tumors reported in autopsy studies of patients of all ages is approximately 0.001%–0.03% (2). The majority of primary cardiac tumors that manifest in children are benign, with rhabdomyoma being the most common neoplasm during fetal life and childhood, followed by teratoma, fibroma, and hemangioma (3). The most common primary malignant pediatric cardiac tumor is sarcoma, including undifferentiated sarcoma, rhabdomyosarcoma, and fibrosarcoma (4).

In recent years, diagnosis of cardiac tumors has been facilitated by the use of noninvasive cardiac imaging. Echocardiography is the first-line modality for evaluation. Cardiac MRI, and to a lesser degree CT, are used for tissue characterization and for evaluation of the size, extension, and physiologic effect (5). In this article, we review common pediatric cardiac and pericardial tumors, emphasizing their imaging characteristics and pathologic features to guide the radiologist and clinicians in diagnosing and managing these lesions.

Benign Cardiac Tumors

Rhabdomyoma

The majority of primary cardiac neoplasms in children are rhabdomyomas (60%), which are congenital hamartomas. Up to 80% of cases are associated with tuberous sclerosis, a genetic disease characterized by the presence of multiple benign hamartomas in several organs (1) (Fig 2). Usually they are discovered prenatally or in early infancy, with a mean patient age at diagnosis of 2 weeks (6,7) (8).

Rhabdomyomas are generally located in the ventricular myocardium or interventricular septum (9); however, they may appear in the atria in up to 30% of cases (10,11). They may be asymptomatic and diagnosed incidentally or may be clinically symptomatic if they protrude into the lumen and obstruct blood flow or promote an accessory pathway that results in cardiac arrhythmia (1,6). Neonates and children may present with murmurs, cyanosis, respiratory distress, cardiac failure, or cardiogenic shock due to myocardial thickening or ventricular dysfunction (2,7,12,13).

Imaging Features.—On echocardiograms, rhabdomyomas appear as a homogeneous, hyperechogenic, solid single mass or as multiple masses of variable sizes (1,6,14). On CT images, rhabdomyomas usually appear as homogeneous, low-attenuation, intramural lesions (Fig 2) and are commonly hypoenhancing after administration of contrast material (10). They appear hyperintense on T2-weighted cardiac MR images and are isointense to slightly hyperintense in comparison with the myocardium at T1-weighted MRI. After intravenous contrast material is administered, they do not demonstrate significant enhancement (15–17) (Fig 3).

Pathologic Features.—Rhabdomyomas are often multifocal (90%) and of variable size, with well-defined borders and a yellow to white color, making them distinct from the surrounding cardiac muscle (Fig 3). They are not encapsulated. Microscopically, these tumors are composed of large polygonal cells with clear vacuoles due to glycogen accumulation. Spider cells are a characteristic feature, named because of the radially oriented myofibrils extending from the nucleus to the cell borders (18).

Treatment and Prognosis.—The majority of rhabdomyomas regress spontaneously in early childhood. Thus, first-line management is often conservative, and surgical resection is reserved for patients with masses that result in hemodynamic disturbances (2,7,14). Alternative clinical treatments have been studied, including everolimus, a mammalian target of rapamycin (mTOR) inhibitor, in patients with clinically significant cardiac rhabdomyomas and tuberous sclerosis disease (1,19). Strong predictors of poor neonatal outcomes include large tumors, hydrops, and fetal dysrhythmias (20).

Fibroma

Cardiac fibromas are the second most common type of benign cardiac tumors in children after rhabdomyomas (10). They are associated with Gorlin syndrome in up to 3% of cases (1,7,10). The onset of symptoms typically occurs later in life, but nearly one-third of patients are diagnosed in utero or during the 1st year of life (21). Most fibromas are solid and solitary and could grow to a size that obliterates the cavity (2,16). They may cause a variety of symptoms due to valvular involvement, obstruction of blood flow, and arrhythmias. Patients may be asymptomatic in one-third of cases or may present with cyanosis, dyspnea, or syncope and may experience sudden death (6). Pericardial effusion

Common Locations of Cardiac Tumors

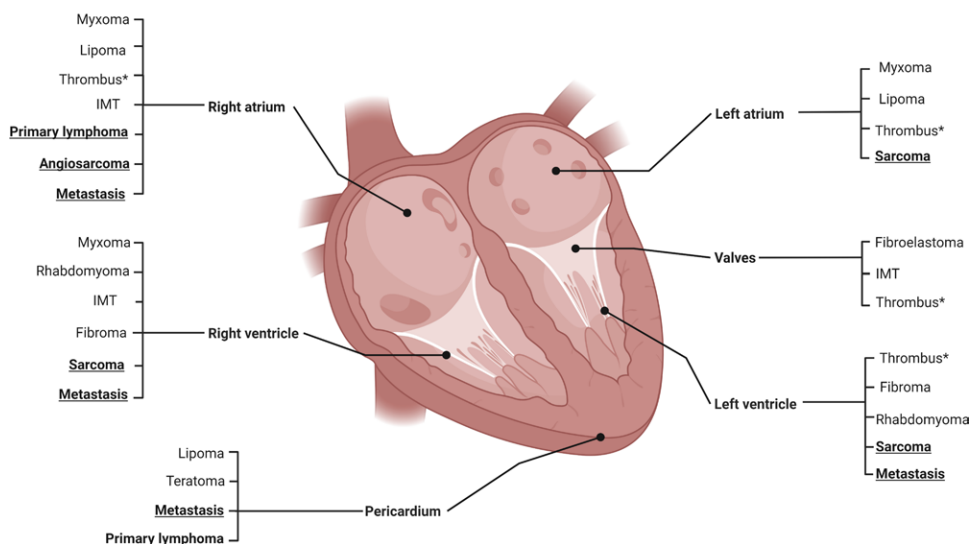


Figure 1. Illustration shows the most common locations of primary cardiac tumors in children. IMT = inflammatory myofibroblastic tumor. * Does not represent a tumor, per se.

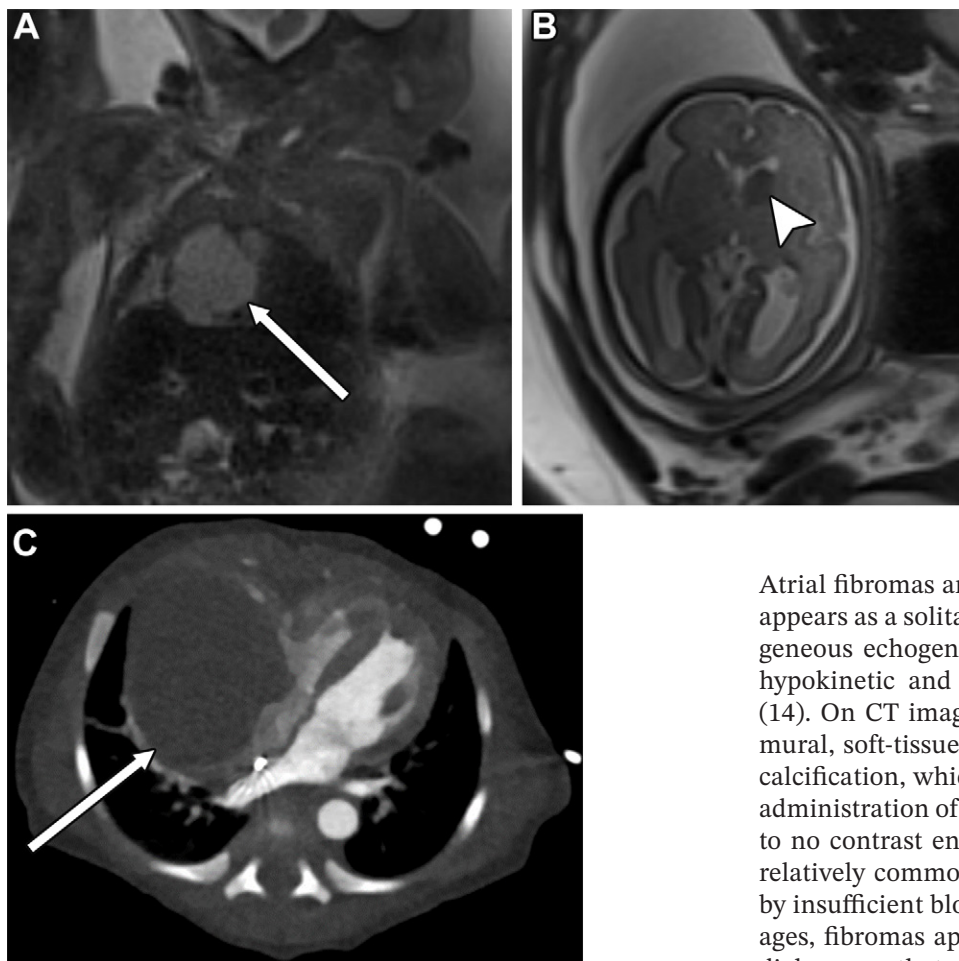


Figure 2. Rhabdomyoma. (A) Fetal coronal T2-weighted MR image shows a large hyperintense lesion arising from the right atrium (arrow). (B) Axial fetal brain T2-weighted half-Fourier rapid acquisition with relaxation enhancement MR image in the same patient shows a T2-hypointense subependymal nodule (arrowhead) and tuberous sclerosis. (C) Axial postnatal contrast-enhanced CT image in the same patient at 1 day old shows a corresponding large hypodense lesion in the right atrium (arrow).

associated with cardiac fibromas has been described in adults (22,23) and children (24).

Imaging Features.—Fibromas are most frequently located in ventricular walls, predominantly in the left ventricle.

Atrial fibromas are rare (6). At echocardiography, a fibroma appears as a solitary, solid, noncontractile mass with heterogeneous echogenicity. The affected myocardium is usually hypokinetic and may mimic a hypertrophic myocardium (14). On CT images, fibromas show a homogeneous, intramural, soft-tissue attenuating mass with or without central calcification, which may be margined or infiltrative. After administration of contrast material, these tumors show little to no contrast enhancement (10). Central calcification is a relatively common and characteristic feature and is caused by insufficient blood supply to the mass. On cardiac MR images, fibromas appear as well-defined solitary intramyocardial masses that are slightly hypointense compared with the adjacent myocardium at T1- and T2-weighted MRI. Gadolinium-enhanced MR angiography shows a hypoperfused mass on first-pass perfusion MR images (25) and strong hyperenhancement on late gadolinium enhancement (LGE) MR images, with or without a hypoenhancing (dark) core (Fig 4). Additional evidence suggests that cardiac MRI with LGE imaging has high accuracy for diagnosis of fibromas, which

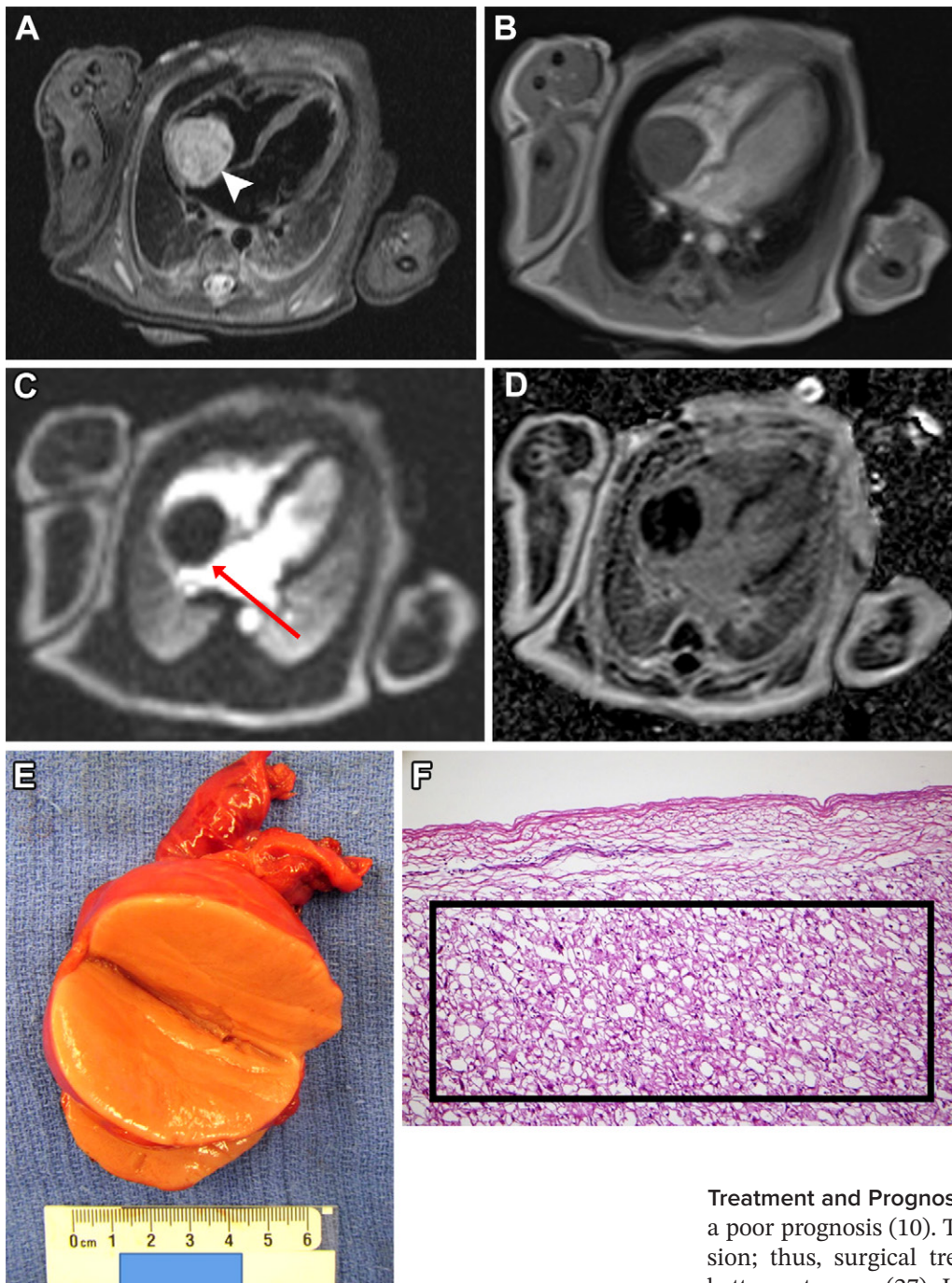


Figure 3. Rhabdomyoma in a 3-day-old male neonate. (A) Axial oblique four-chamber-view T2-weighted MR image with fat saturation shows a T2-hyperintense lesion in the right atrium (arrowhead). (B) Axial oblique steady-state free-precession (SSFP) bright-blood MR image shows mild hypointensity of the lesion relative to the normal myocardium. (C) Axial oblique contrast-enhanced first-pass fusion MR image shows no enhancement of the lesion (arrow). (D) Axial oblique late gadolinium-enhancement (LGE) MR image shows no substantial contrast enhancement. (E) Gross photograph shows a well-circumscribed firm tumor with a glistening cut surface. (F) Photomicrograph shows a heterogeneous population of rhabdoid cells (rectangle), some with vacuolated cytoplasm and small basophilic nuclei. (Hematoxylin-eosin [H-E] stain; original magnification, $\times 100$.)

may enable clinicians to nearly forgo the use of diagnostic biopsy for diagnosis (24).

Pathologic Features.—Cardiac fibromas have a predilection for the ventricular septum, although they may be seen in other locations. The lesions grossly resemble leiomyomas, appearing well circumscribed, with a firm white whorled cut surface (Fig 4). Bland spindle cells (fibroblasts) are present in a collagenous background at histologic examination. Younger patients often have more cellular lesions, which decrease with age in concert with increased collagen. Focal calcification and elastic fibers may be seen (26).

Treatment and Prognosis.—In general, cardiac fibromas have a poor prognosis (10). They do not show spontaneous regression; thus, surgical treatment is recommended to achieve better outcomes (27). Location, size, and surgical resectability of fibromas determine treatment, which may be complete surgical resection, incomplete excision, or an orthotopic heart transplant for large tumors (1,6,10).

Inflammatory Myofibroblastic Tumor

Inflammatory myofibroblastic tumor (IMT) is a rare mesenchymal neoplasm predominantly occurring in children, adolescents, and young adults (28). Cardiac involvement is rare, with only approximately 60 cases reported in the scientific literature. It usually manifests as one or more endocardial-based intracavitary masses and most commonly involves the valve leaflets (1,29). These tumors may be mobile and are at risk for extending through the atrioventricular valves or outflow tracts to the great arteries (29).

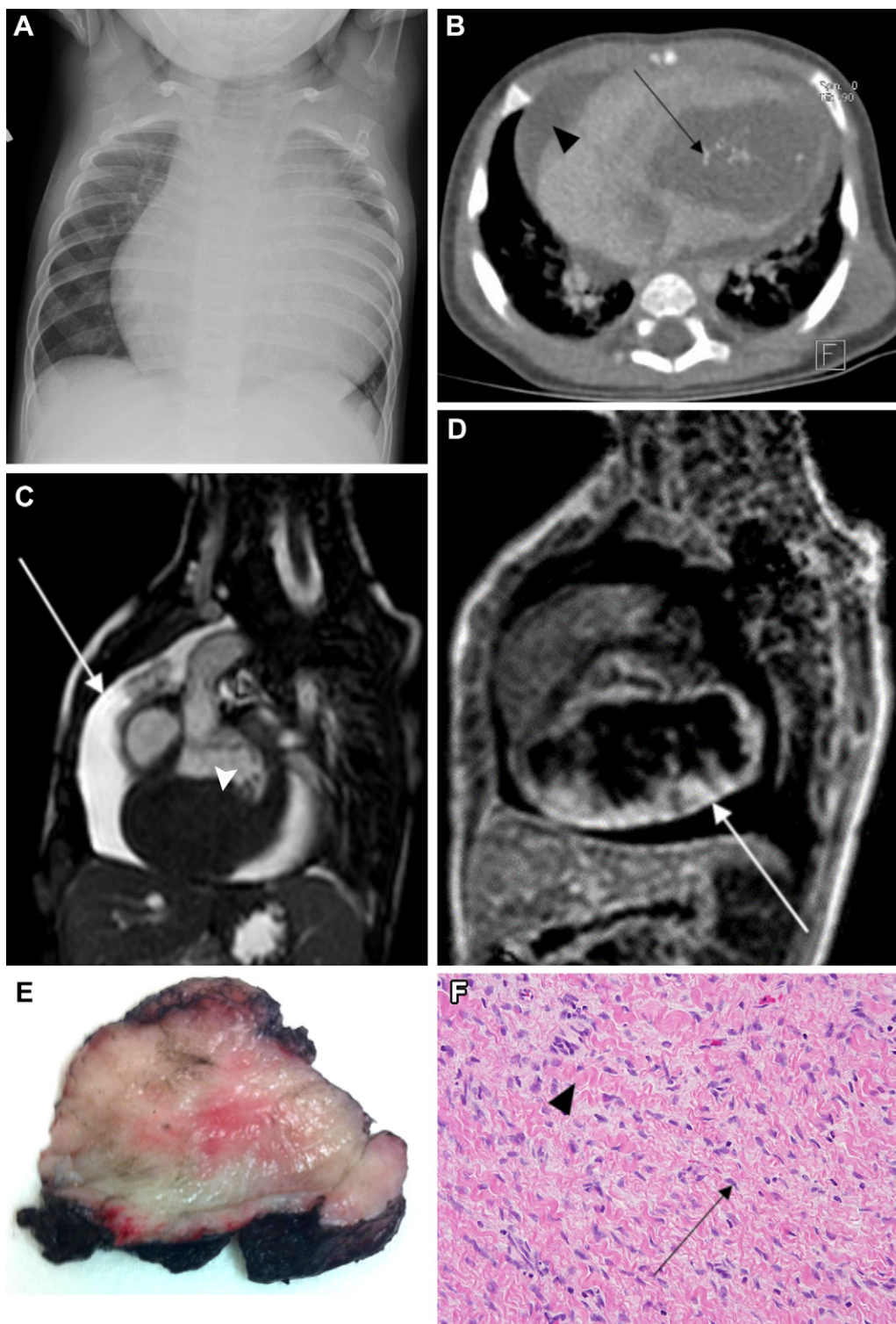


Figure 4. Fibroma in a 9-month-old male infant. (A) Frontal chest radiograph shows an enlarged cardiomeastinal silhouette. (B) Contrast-enhanced axial CT image of the chest shows a large hypo-enhancing mass extending into the left ventricle, with internal calcifications (arrow) and a pericardial effusion (arrowhead). (C) Sagittal SSFP bright-blood cardiac MR image shows a homogeneous hypointense intramyocardial lesion (arrowhead) along the free wall of the left ventricle and a large pericardial effusion (arrow). (D) Sagittal contrast-enhanced LGE MR image shows the heterogeneous appearance of the ventricular mass, with avid peripheral enhancement (arrow) and a central hypointense core. (E) Gross photograph shows a poorly circumscribed soft-tissue mass with a central area of firmness. (F) High-power photomicrograph shows proliferation of bland, spindle, and elongated fibroblasts (arrow) intermixed in a heavily collagenized stroma (arrowhead) with no mitoses. (H-E stain; original magnification, $\times 100$.)

Symptoms are typically based on the location of the tumor, although up to 30% of patients may have constitutional symptoms, including localized pain, fever, weight loss, and malaise (28). Patients may also present with shortness of breath, syncope, chest pain, transient ischemic attacks, or embolic events (1).

Imaging Features.—The imaging appearance of IMT may be affected by the ratio of inflammatory to fibroblastic components

and the presence of any myxoid elements (1). These tumors are typically lobulated, with a broad-based attachment. At echocardiography, IMTs are heterogeneously hyperechoic, with some vascularity (30). They may be mobile at cine imaging, and pedunculated lesions are at risk for prolapse depending on their location (31). CT shows a lobulated broad-based mass that may show patchy contrast enhancement. On fluorine 18 (^{18}F) fluorodeoxyglucose (FDG) PET/CT, FDG uptake can vary from low to high depending on tumor cellularity and biologic behavior

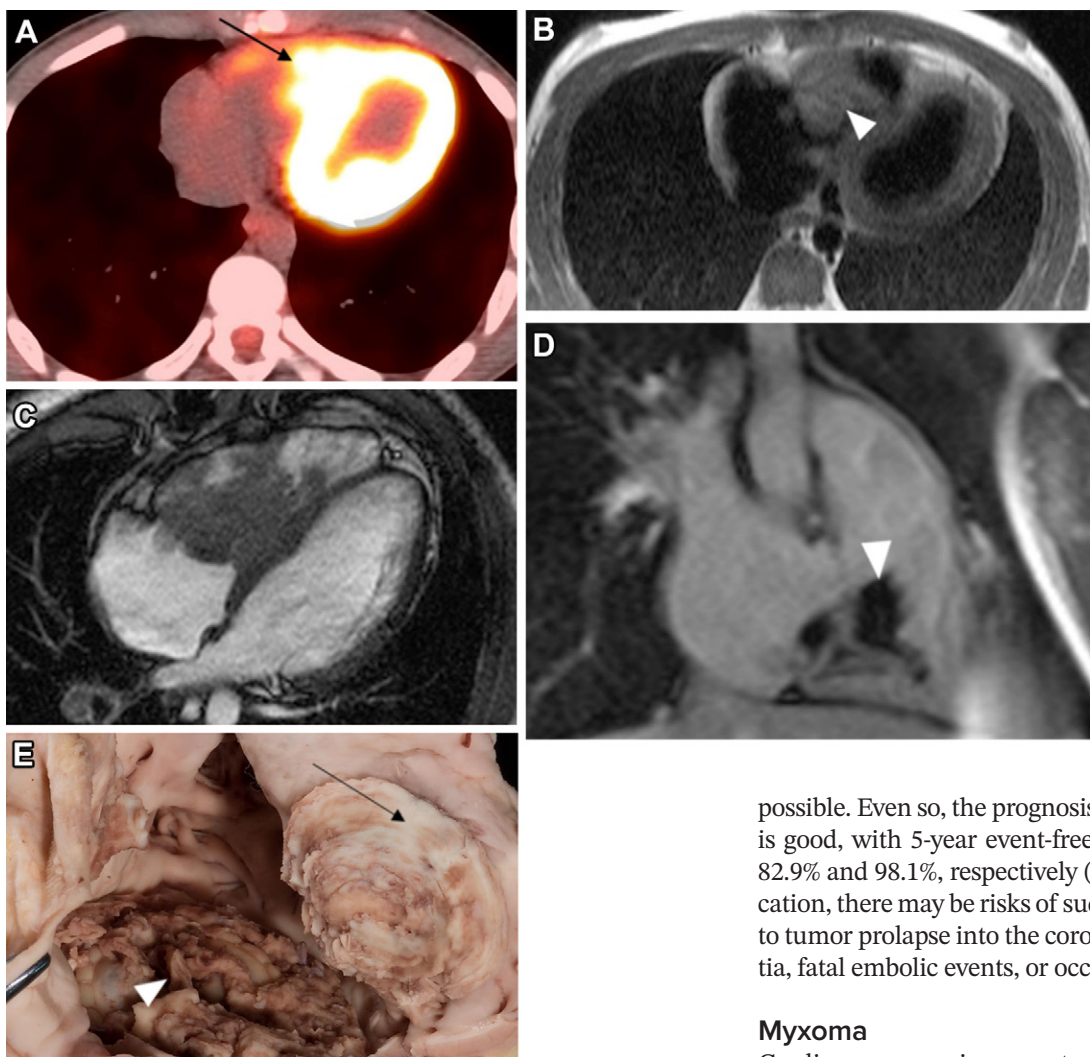


Figure 5. IMT in a 15-year-old adolescent boy. (A) Axial PET/CT image shows hypermetabolic activity in the right ventricle corresponding to the mass (arrow). (B, C) Axial dark-blood MR image (B) and SSFP bright-blood four-chamber projection MR image (C) show a lobulated broad-base mass in the right ventricle (arrowhead in B) that is isointense to the myocardium. (D) Oblique postcontrast true fast imaging with steady-state precession (TrueFISP) shows extension across the tricuspid valve, as well as a very low-signal component of the mass, consistent with thrombus (arrowhead). (E) Magnified image of the gross specimen along the junction of the right atrium and right ventricle shows a more fibrous and well-defined tumor component (arrow) as well as a more fungating darker component mixed with thrombus (arrowhead).

of the tumor cells (32). The appearance at cardiac MRI varies depending on the composition of the IMTs. They can be isointense in comparison with the myocardium at T1-weighted MRI (Fig 5) and hypointense at T2-weighted MRI if there are relatively more fibrous components, but they may show high T2 signal intensity if they contain myxoid elements (1,30,31). There may be persistent enhancement during the delayed phase at CT and intense delayed enhancement at cardiac MRI (1).

Pathologic Features.—IMT typically appears as a polypoid mass originating in the endocardium (Fig 5). The myofibroblastic cells of the tumor may be spindle or plump and may be set in a background containing variable amounts of myxoid stroma and chronic inflammatory cells (eg, lymphocytes, plasma cells, and eosinophils). A thin layer of fibrin may be present on the surface (33).

Treatment and Prognosis.—Surgical resection is the treatment of localized disease. No known cases of metastasis have been reported with cardiac IMT, although rare occurrences of metastatic disease have been reported with IMT at other sites. There is an increased risk of recurrence if complete resection is not

possible. Even so, the prognosis for patients with IMT in all sites is good, with 5-year event-free survival and overall survival of 82.9% and 98.1%, respectively (34). Depending on the tumor location, there may be risks of sudden death with cardiac IMT due to tumor prolapse into the coronary arteries via the coronary ostia, fatal embolic events, or occlusion of outflow tracts (29).

Myxoma

Cardiac myxoma is a rare tumor in children, but it deserves special attention given its association with Carney complex and its characteristic imaging appearance. Carney complex is a rare familial multiple endocrine neoplasia syndrome caused by mutations of the *PRKARIA* gene, and cardiac myxomas are the most common noncutaneous diagnostic criteria for this complex (35,36). When they occur in these patients, cardiac myxomas are often multicentric and more aggressive and may be metachronous (36–38). Other tumors associated with this complex that may be detected at imaging include primary pigmented nodular adrenocortical disease, large-cell calcifying Sertoli cell tumors, psammomatous melanotic schwannomas, and thyroid nodules (36).

Imaging Features.—When sporadic, cardiac myxomas almost always occur in the left atrium with a characteristic stalk-like attachment to the interatrial septum near the fossa ovalis. They may be mobile and prolapse across the mitral valve during diastolic filling. However, cardiac myxomas in patients with Carney complex can occur in any chamber and tend to be multiple.

Calcification may be visible on radiographs, and when located in the left atrium, these tumors may result in other radiographic signs of valvular obstruction such as pulmonary edema. Echocardiography shows a heterogeneous mass, occasionally with suggestion of calcification. At CT, they are

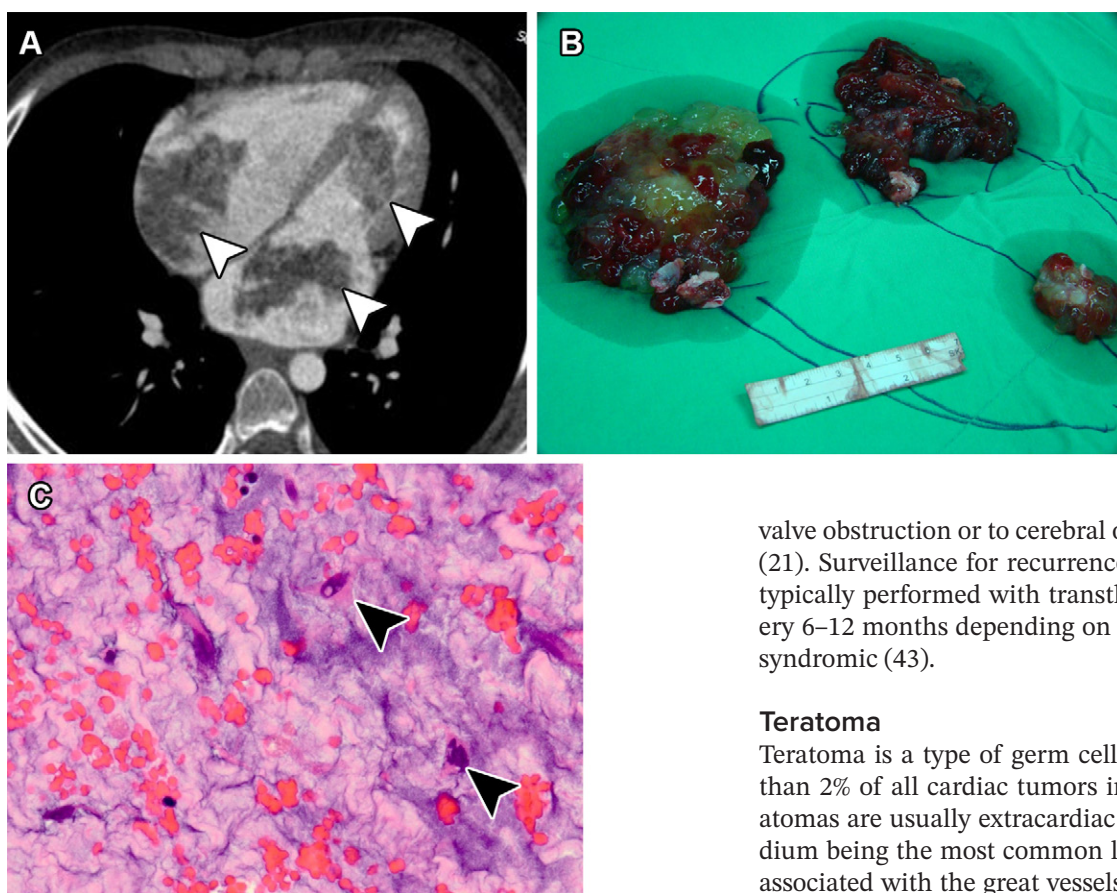


Figure 6. Myxoma in a 14-year-old adolescent girl. (A) Axial contrast-enhanced portal venous phase CT image shows three pedunculated masses with irregular borders within the right and left atrium and left ventricle (arrowheads). (B) Gross photograph of the three masses shows a lobulated and gelatinous appearance. (C) High-power photomicrograph shows scattered polygonal cells (arrowheads) within a myxoid homogeneous stroma. (H-E stain; original magnification, $\times 400$.)

typically spherical or ovoid and occasionally lobulated, lower in attenuation than surrounding myocardium with heterogeneous enhancement (Fig 6) (39) (40).

On cardiac MR images, cardiac myxomas are typically hyperintense at T2-weighted MRI, show relative hypoperfusion to myocardium at first-pass perfusion imaging, and show heterogeneous enhancement at LGE imaging. Balanced steady-state free-precession (SSFP) MR imaging may show the mobility of the mass and the contact with the valvular structures of the outflow tracts (41).

Pathologic Features.—Myxomas are heterogeneous neoplasms of uncertain histogenesis. They can be pedunculated or sessile, with a wide range in size. The firm to gelatinous mass may have focal areas of hemorrhage (Fig 6). Although myxomas are typically smooth surfaced, excrescences may be present and are more likely lead to a tumor embolus. Histologically, characteristic myxoma cells, also known as lepidic cells, contain an eosinophilic cytoplasm and bland round to oval nuclei, with inconspicuous nucleoli. These cells can be arranged in nests around vascular channels (ie, perivascular rings) or in a myxoid background, typically infiltrated by varying numbers of inflammatory cells. Recent or remote hemorrhage is frequently identified (42).

Treatment and Prognosis.—Cardiac myxomas in patients with Carney complex account for more than 50% of disease-specific mortality secondary to complete mitral valve or tricuspid

valve obstruction or to cerebral or coronary embolization (36) (21). Surveillance for recurrence or metachronous disease is typically performed with transthoracic echocardiography every 6–12 months depending on whether a case is sporadic or syndromic (43).

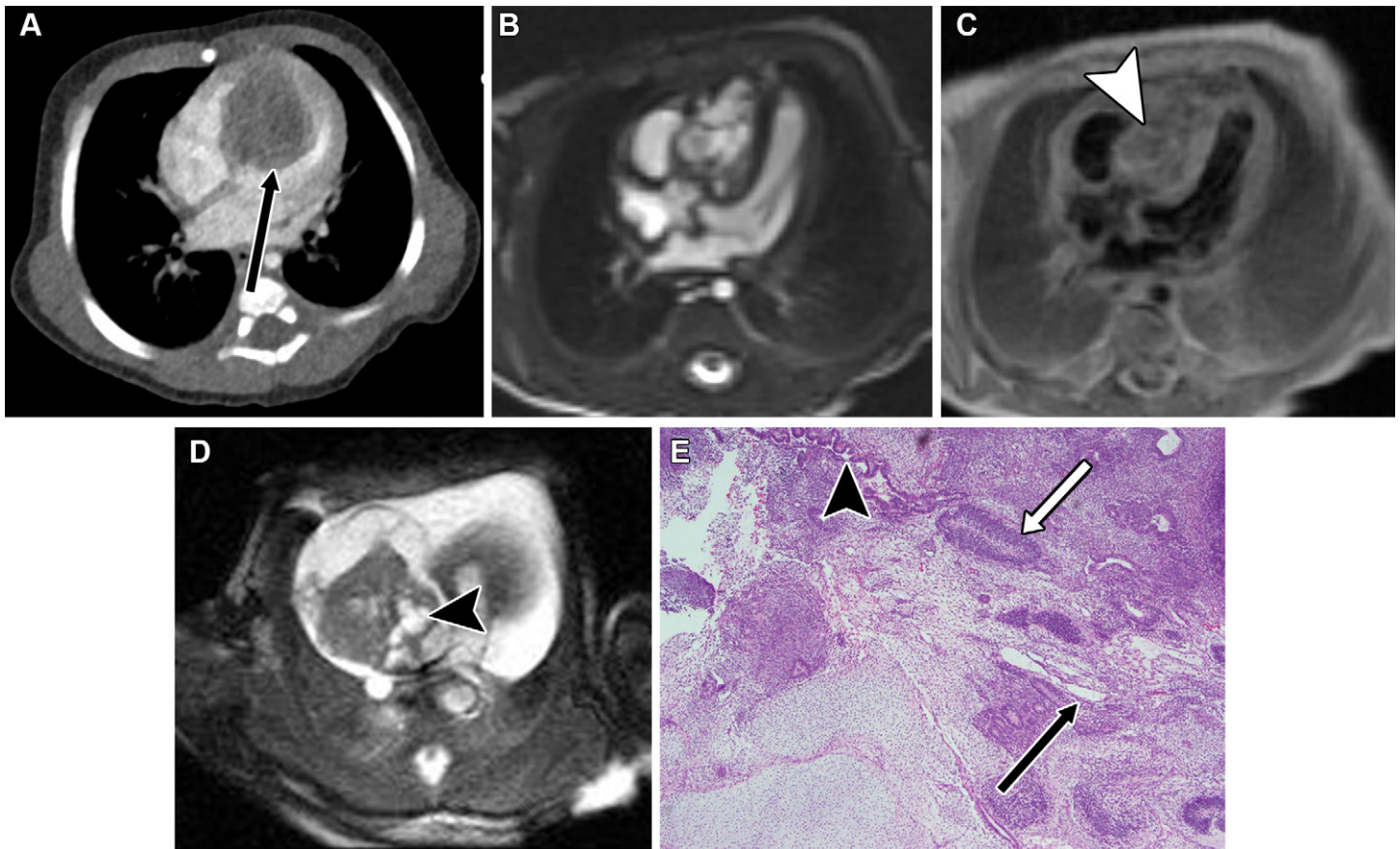
Teratoma

Teratoma is a type of germ cell tumor and accounts for less than 2% of all cardiac tumors in pediatric patients (44). Teratomas are usually extracardiac in location, with the pericardium being the most common location, and they are usually associated with the great vessels, most often arising from the root of the ascending aorta (45). They rarely involve the myocardium, manifesting as an intracardiac mass arising from the atrial or ventricular wall (46). Greater than 75% of cardiac teratomas occur in children aged 15 years or younger (25).

The patient presentation can be varied, with symptoms usually attributable to the mass effect from the tumor on the heart and potential symptoms including arrhythmia, heart failure, valvular insufficiency, hydrops, or in rare cases, sudden death; neonates may die at birth from cardiac tamponade and cardiac compression (3,47).

Imaging Features.—Chest radiography may show a mass in the aortopulmonary window (48), an enlarged cardiomeastinal silhouette, and in rare cases, calcification or formed teeth (21). Teratomas are usually easily detected at fetal echocardiography or postnatal echocardiography as a heterogeneous and encapsulated cystic mass, typically positioned on the right side of the heart and frequently associated with pericardial effusion (45,48). CT findings are of a heterogeneous tumor that may show areas of hypoattenuation representing fat or hyperattenuating areas of calcifications, which can be helpful in supporting the diagnosis (40). On cardiac MR images, teratomas are described as multilocular tumors, with cystic components and intervening solid areas, typically displacing the heart and rotating it along its longitudinal axis (25). The tumor is usually intrapericardial with mass effect on the superior vena cava or right atrium. A heterogeneous cystic mass, isointense or hypointense to the myocardium at T1-weighted MRI and with heterogeneous signal intensity at T2-weighted MRI, is often seen, with areas

Figure 7. Teratoma in two patients. (A–C) Axial contrast-enhanced portal venous phase CT image (A) in a 3-day-old female neonate shows a hypointense mildly heterogeneous mass in the right ventricle (arrow). Axial SSFP MR image (B) shows a heterogeneous hyperintense mass in the same location. Axial LGE postcontrast MR image (C) shows signal hypointensity, with scattered areas of minimal enhancement (arrowhead). (D, E) Oblique axial SSFP MR image (D) in a different patient shows a large heterogeneous mass with solid and cystic components (arrowhead) and a large pericardial effusion. Low-power photomicrograph (E) shows focal areas with mature components (cartilage [black arrow] and columnar epithelium [arrowhead] adjacent to tubules lined by immature neuroepithelium [white arrow]). (H-E stain; original magnification, $\times 20$.)



of high signal intensity at T2-weighted MRI corresponding to cysts (Fig 7). Teratomas may be isointense or hypointense to the myocardium at first-pass perfusion imaging and can show heterogeneous enhancement with LGE (1,49).

Pathologic Features.—Pericardial teratomas are most frequently a single mass with a wide range of sizes, located at the base of the heart. The cut surface is lobulated, with cystic and solid areas. Microscopically, tissues representing all three germ layers are present (Fig 7). Immature components are especially common in infants and bear no prognostic significance in the absence of a somatic malignancy. In rare cases, malignant components have been described (50).

Treatment and Prognosis.—Surgical excision is the only effective treatment of a cardiac teratoma (25,51). As expected, intracardiac teratomas are more difficult to remove due to their interventricular septal location. Total or partial surgical resection usually has a favorable outcome (45). Pericardiocentesis or fetal surgery may be performed to excise the tumor in utero (3,46).

Congenital Hemangioma

Congenital hemangiomas can occur anywhere in the heart; however, the most common sites are the right atrium and the

interventricular septum (3,6). Although these tumors are usually found incidentally, they can cause hydrops and cardiac tamponade secondary to large pericardial effusions in infants. The presence of additional hemangiomas in the skin should raise the suspicion for a hemangioma in a child with a cardiac lesion (3).

Imaging Features.—Hemangiomas have variable locations; their presence has been reported in the atria and ventricles. Some are mobile. At echocardiography, they are solid and hyperechoic masses. At angiography, there is a characteristic “blush” as contrast material enhances the mass vascularity. On CT images, they appear as heterogeneous lesions that avidly enhance and can contain calcifications (8). These masses are isointense to the myocardium on T1-weighted MR images and hyperintense on T2-weighted MR images. Contrast-enhanced first-pass perfusion MR imaging shows avid enhancement (Fig 8). Hemangiomas are isointense to the myocardium to hyperintense at cine SSFP MR imaging and often have a heterogeneous appearance at delayed enhancement MRI.

Pathologic Features.—Cardiac hemangiomas may be either intramyocardial or endoluminal, with endoluminal lesions showing a sessile or polypoid appearance on gross examination.

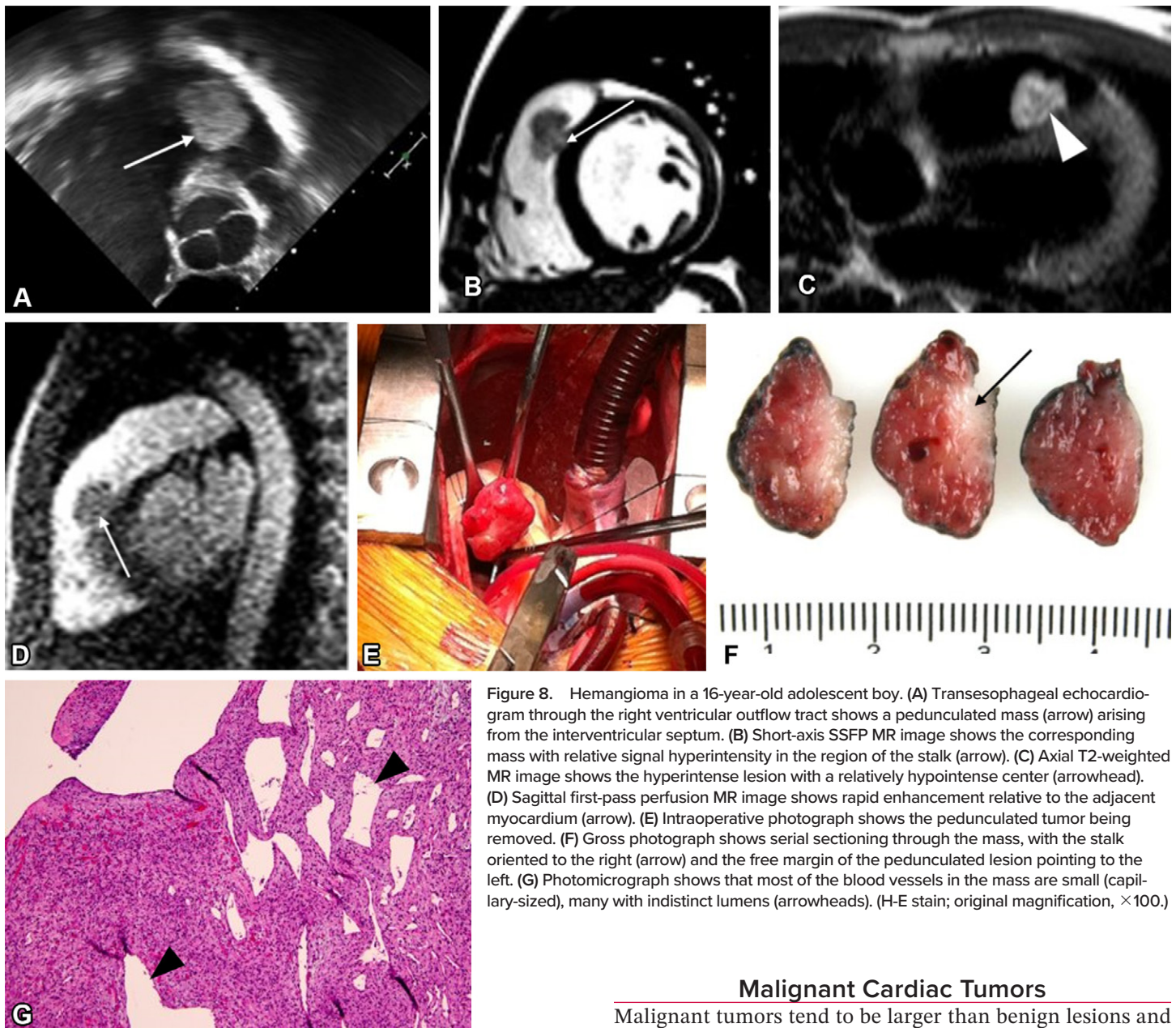


Figure 8. Hemangioma in a 16-year-old adolescent boy. (A) Transesophageal echocardiogram through the right ventricular outflow tract shows a pedunculated mass (arrow) arising from the interventricular septum. (B) Short-axis SSFP MR image shows the corresponding mass with relative signal hyperintensity in the region of the stalk (arrow). (C) Axial T2-weighted MR image shows the hyperintense lesion with a relatively hypointense center (arrowhead). (D) Sagittal first-pass perfusion MR image shows rapid enhancement relative to the adjacent myocardium (arrow). (E) Intraoperative photograph shows the pedunculated tumor being removed. (F) Gross photograph shows serial sectioning through the mass, with the stalk oriented to the right (arrow) and the free margin of the pedunculated lesion pointing to the left. (G) Photomicrograph shows that most of the blood vessels in the mass are small (capillary-sized), many with indistinct lumens (arrowheads). (H-E stain; original magnification, $\times 100$.)

Histologically, cardiac hemangiomas are similar to those in other sites and can also change in appearance over time. Proliferating lesions contain numerous capillaries lined by bland-appearing endothelial cells, some with indistinct lumens (Fig 8). Mitotic figures are often easily identified in these earlier lesions. In cases of infantile hemangioma, expression of GLUT-1 can be seen in the cytoplasm of endothelial cells with immunohistochemical staining, allowing them to be distinguished from other vascular neoplasms and malformations. However, congenital hemangiomas lack expression of GLUT-1 (52).

Treatment and Prognosis.—The treatment of choice for hemangiomas is surgical excision, although spontaneous regression has been reported in the literature (53). Long-term outcomes after surgical resection are excellent.

Malignant Cardiac Tumors

Malignant tumors tend to be larger than benign lesions and irregularly shaped, and they invade surrounding tissues (54).

Primary Cardiac Sarcoma

Primary cardiac sarcomas represent the most common primary malignant tumor in adults and children, accounting for 75% of primary malignant cardiac masses (1). Undifferentiated sarcomas, rhabdomyosarcomas, and fibromyosarcomas are the most common. Nonetheless, giant cell sarcomas, synovial sarcomas, leiomyosarcomas, myeloid sarcomas, and osteosarcomas can also be seen (4).

Sarcomas can manifest anywhere in the heart, but some lesions have a higher preference for the left atrium, including undifferentiated sarcomas (9,55) (25). Rhabdomyosarcoma is the most common primary cardiac malignancy of childhood. Although it has no predilection for any cardiac chamber, it more frequently involves the cardiac valves compared with other sarcomas. The clinical presentation of sarcomas is variable, with symptoms including recurrent pericardial

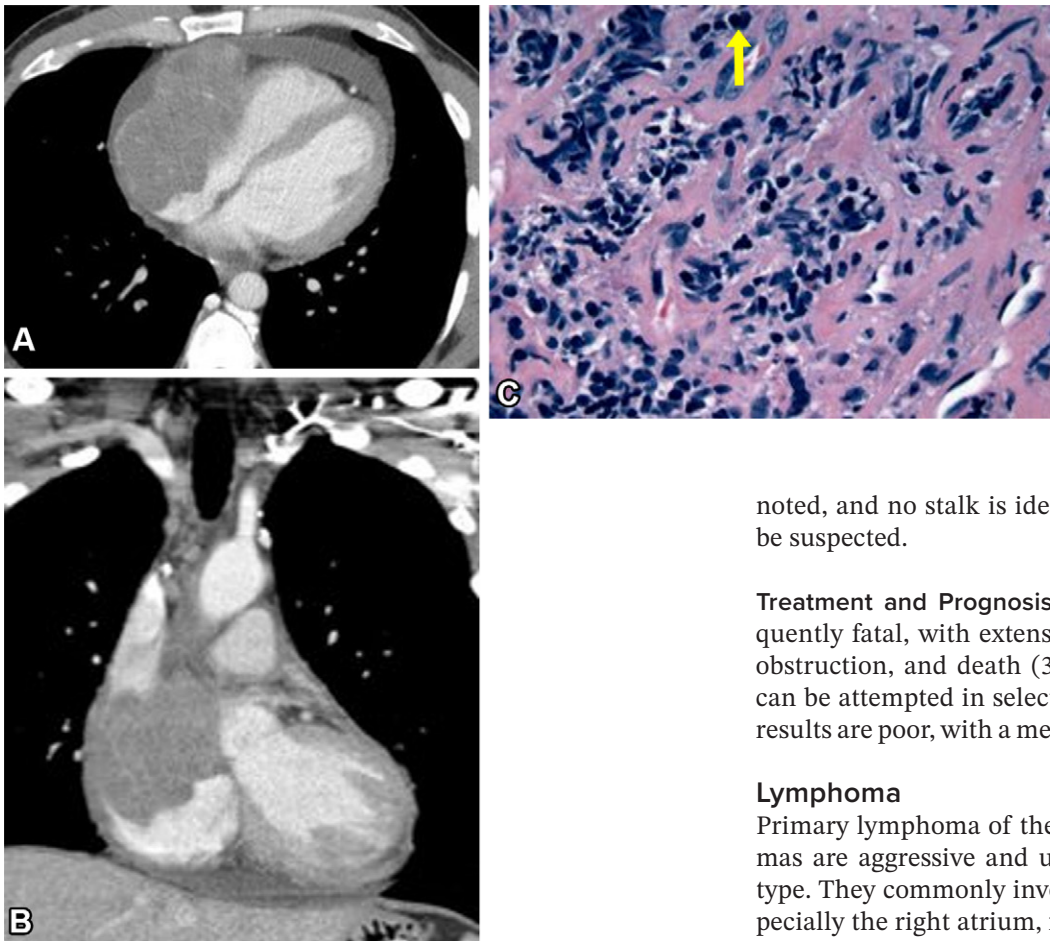


Figure 9. Large B-cell lymphoma. (A, B) Axial (A) and coronal (B) contrast-enhanced portal venous phase CT images show a large mildly heterogeneous hypoattenuating mass arising from the right atrium, with minimal contrast enhancement. (C) Photomicrograph shows malignant cells with an increased nuclear to cytoplasmic ratio (arrow). (H-E stain; original magnification, $\times 400$.)

effusion, chest pain, respiratory distress, pulmonary emboli, arrhythmia, or congestive heart failure.

Imaging Features.—At echocardiography, sarcomas appear as a heterogeneous broad-based mass, with areas of hypoechogenicity that may indicate tumor necrosis or hemorrhage. When present, associated pericardial effusion suggests pericardial infiltration (9). On cardiac MR images, sarcomas demonstrate an infiltrative appearance defined as crossing a cardiac annular or tissue plane, involving both cardiac and extracardiac structures, with possible linear growth through a large vessel. None of the cardiac MRI tissue characterization sequences are sensitive or specific for the diagnosis of a malignant lesion, with a variable appearance at cine SSFP, T1-weighted, and T2-weighted MRI. However, an infiltrative appearance is strongly associated with a malignant cardiac tumor (24,56).

Pathologic Features.—Sarcomas are usually large infiltrative firm white masses, although calcification and/or hemorrhage may impart textural or color changes. A wide range of microscopic morphology is possible depending on the histologic type, including fibrosarcoma, rhabdomyosarcoma, liposarcoma, and leiomyosarcoma, among others (57). Because of their location, left atrial sarcomas can mimic the appearance of left atrial myxomas. If invasive features are shown, heterogeneous areas of necrosis and hemorrhage are

noted, and no stalk is identified, a primary sarcoma should be suspected.

Treatment and Prognosis.—The prognosis is poor and frequently fatal, with extensive local infiltration, intracavitary obstruction, and death (3). Aggressive surgical approaches can be attempted in selected patients; however, the surgical results are poor, with a mean survival time of 11 months (58).

Lymphoma

Primary lymphoma of the heart is extremely rare. Lymphomas are aggressive and usually of the non-Hodgkin B-cell type. They commonly involve the right side of the heart, especially the right atrium, followed by the right ventricle and left ventricle (45). Patients can present with dyspnea and chest pain; however, specific locations can produce certain clinical syndromes such as superior vena cava syndrome, with right atrial lymphoma. Arrhythmia, coronary artery obstruction, tumor embolism, and pericardial effusion can also be seen with tumor infiltration (8).

Imaging Features.—At echocardiography, lymphomas can appear as localized wall thickening secondary to homogeneous infiltrating masses or as nodular masses protruding into the cardiac chambers. In cases where the atrioventricular groove is affected, there can be encasement of the right coronary artery and invasion of the pericardium, with resulting effusion (9). Primary cardiac lymphoma typically is a homogeneously low-attenuating hypoenhancing mass on contrast-enhanced CT images (Fig 9). On cardiac T1-weighted MR images, it appears as signal isointensity to slight hypointensity and as isointensity on T2-weighted MR images compared with the myocardium. There is minimal contrast material uptake on LGE MR images (15).

Pathologic Features.—Lymphoma displays a firm, white-tan, fleshy, lobulated cut surface on gross inspection. Most reported cases correspond to diffuse large B-cell lymphoma, although many other types of lymphoma (eg, Burkitt lymphoma) and immunodeficiency-associated lymphoproliferative disorders may be diagnosed. Microscopically, diffuse large B-cell lymphomas are composed of large atypical lymphoid cells (Fig 9). Expression of B-cell markers such as

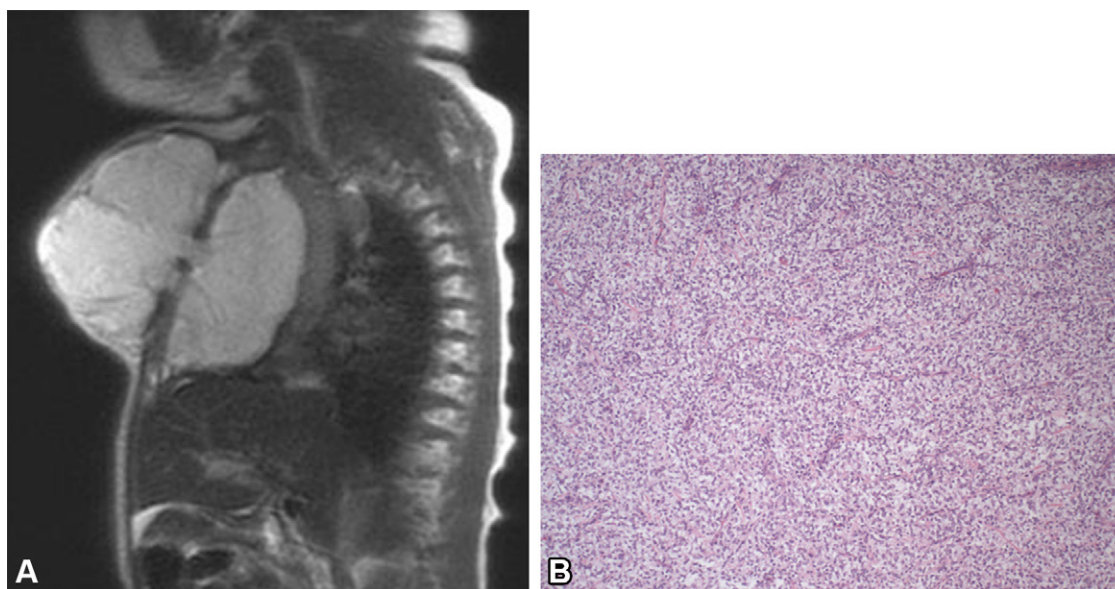


Figure 10. Infantile fibrosarcoma of the chest wall extending into the heart in a 3-day old neonate (A) Sagittal T2-weighted MR image shows a large hyperintense infiltrating lesion arising from the anterior chest wall, invading the mediastinum. (B) Low-power photomicrograph shows a poorly differentiated tumor with proliferation of small- to medium-sized spindled cells in a myxoid background, with mild nuclear atypia and a vague fascicular pattern. (H-E stain; original magnification, $\times 40$.)

CD19, CD20, CD22, CD79a, or PAX-5 may be demonstrated at immunohistochemical analysis or flow cytometry (59).

Primary cardiac lymphomas are usually uniform tumors. Necrosis and involvement of the cardiac valves are atypical and suggest an alternative diagnosis such as angiosarcoma, especially if they manifest in the right atrium. Lymphomas are typically homogeneous, centered on the atrioventricular groove and sparing the right coronary artery. Angiosarcomas are usually centered on the right atrial free wall and show a heterogeneous infiltrative appearance, with areas of necrosis or hemorrhage.

Treatment and Prognosis.—While the prognosis is overall poor, early diagnosis and institution of chemotherapy may result in remission of disease and symptom relief, with longer survival time (21). Radiation therapy and surgical resection can be considered; however, they are less favorable treatments.

Metastatic Tumors

Metastatic cardiac tumors are more common than primary cardiac tumors (3). In children, cardiac metastases can occur from lymphoma, sarcoma, testicular carcinoma, and Wilms tumor (1). Metastatic lesions are generally seen late in the course of a primary tumor and may arise anywhere in the heart or may diffusely extend along the epicardial or pericardial surface (Fig 10), with morphology dependent on the primary cell type (12). Patients usually present with multiple solid nodules associated with adhesions and hemorrhagic effusions. In addition, intracardiac lesions can be seen with direct extension of primary malignancies such as Wilms tumor via the renal veins, through the inferior vena cava, into the right atrium.

Imaging studies are an invaluable tool in the diagnosis of cardiac metastatic disease. If a patient with a known malignancy

presents with new cardiovascular symptoms, cardiac involvement should be included in the differential diagnosis and echocardiography should be performed as the initial diagnostic test. A chest radiograph could show an enlarged cardiac silhouette if a pericardial effusion is present. Cardiac CT and cardiac MRI can be subsequently performed to evaluate for intramyocardial masses and direct tumor extension.

In recent studies of adult cardiac tumors, the presence of increased metabolic activity in cardiac lesions at FDG PET/CT has been associated with an increased likelihood of malignancy (60). However, nonspecific FDG uptake in the normal myocardium is a common finding that can range from uniform and intense to irregular or absent and can represent a diagnostic challenge.

Pericardial Lesions

Lipoma and Lipoblastoma

Fatty lesions originating in the pericardium are most commonly lipomas. Lipomas are fatty soft-tissue masses encapsulated with a thin layer of fibrous tissue that can occur at any age, including in a fetus (61). Pericardial lipomas are less circumscribed than are subcutaneous lipomas, often conforming to the space that they occupy (62).

Conversely, pericardial lipoblastomas are less common than lipomas and are composed of fetal adipose tissue. These tumors most frequently occur in patients younger than 3 years old (63). Superficial lipoblastomas are usually well encapsulated, but a more diffuse infiltrative form can be present, which is classified as lipoblastomatosis (64).

Imaging Features.—Pericardial lipomas and lipoblastomas have similar imaging findings. Both are echogenic at echo-

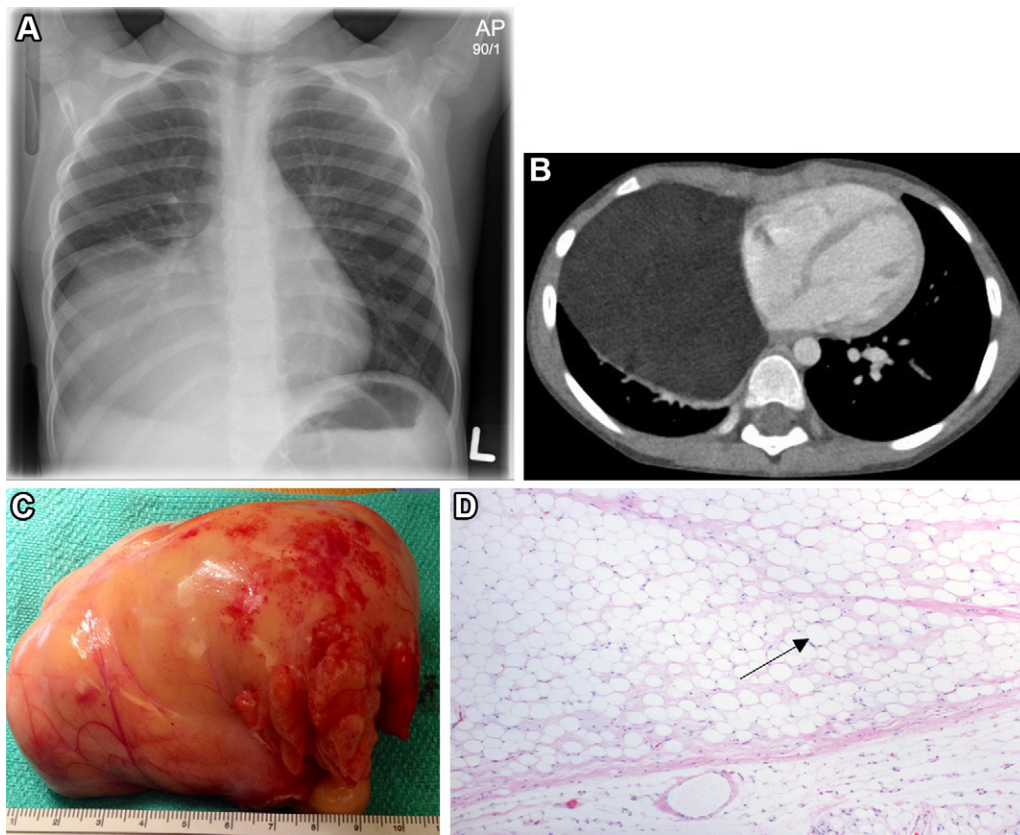


Figure 11. Pericardial lipoma in a 14-year-old adolescent girl. (A) Anteroposterior chest radiograph shows a large soft-tissue attenuating mass along the right cardiac border. (B) Corresponding contrast-enhanced axial CT image shows a large fat-attenuation lesion arising from the right pericardial margin. (C) Gross photograph shows a lobulated and encapsulated soft-tissue mass. (D) Photomicrograph shows mature adipocytes (arrow) consistent with lipoma. (H-E stain; original magnification, $\times 10$.)

cardiography. At CT, they appear as a nonenhancing low-attenuation (less than 0 HU) lesion (64) (Fig 11). Lipomas are hyperintense at T1-weighted MRI and show uniform fat saturation and no first-pass perfusion after administration of gadolinium-based contrast material (56). These lesions may or may not have a visible capsule on images (64). Lipoblastomas sometimes have intrinsic stranding or swirls at CT or MRI that correspond to their fibrovascular network (62). In addition, cystic changes may occasionally be present in lipoblastomas (64).

Pathologic Features.—Lipomas are well-circumscribed, yellow, lobulated masses. They usually measure 2–10 cm, although those in a deep intramuscular location tend to be larger. Microscopically, they are composed of bland mature adipocytes arranged in sheets (Fig 11). Some can manifest with degenerative changes such as calcifications, hemorrhage, or necrosis (65).

Lipoblastomas display a soft lobulated yellow, white, or tan surface grossly. They might also show areas with myxoid change. Microscopically, they consist of lobulated sheets of adipocytes, revealing a spectrum of maturation ranging from primitive stellate or spindled mesenchymal cells to multivacuolated or small signet ring cell lipoblasts with fibrous septa dividing the tumor into lobules and a plexiform vasculature composed of thin-walled vessels (63).

Treatment and Prognosis.—Lipomas may be treated conservatively or with resection, but children, especially those younger than 3 years old who have a rapidly enlarging mass,

should undergo resection given the differential diagnosis consideration of lipoblastoma (66). Recurrence after resection of a cardiac lipoma is rare but has been reported (67). Surgical resection of lipoblastomas is the treatment of choice, but complete resection is essential to avoid recurrence. Up to 80% of incompletely resected lipoblastomas recur (63). Follow-up imaging after resection of a lipoblastoma for up to 5 years is recommended (66).

Cardiac Tumor Mimics

Thrombus

The heart is the most common site for thrombosis in children, representing approximately 25%–28% of cases, most commonly associated with indwelling central venous catheters, followed by locations in the peripheral arteries, central nervous system vessels, major intra-abdominal veins, and peripheral veins (68). Intracardiac thrombus is the main consideration in the differential diagnosis when a cardiac mass is encountered, and it can occur anywhere in the heart.

Echocardiography is the primary modality for evaluation of a thrombus, which can be seen as a laminar and adherent mural mass or a nonpedunculated and mobile intraluminal mass with heterogeneously increased echogenicity (69). At CT, thrombi show homogeneous attenuation and can contain calcifications in patients with a chronic thrombus. On cardiac MR images, the typical appearance is a hypointense nonenhancing mass. The specific imaging characteristics depend on the age of the thrombus. An acute thrombus is generally hyperintense on T1- and T2-weighted MR images; a subacute thrombus is

hyperintense on T1-weighted MR images and hypointense on T2-weighted MR images and shows homogeneously low signal intensity on gradient-echo MR images. Organized chronic thrombus is hypointense on T1- and T2-weighted MR images and shows intermediate signal intensity on gradient-echo MR images (45). On LGE MR images with a long inversion time (600 msec), a thrombus shows low signal intensity, and a tumor may show an increase in signal intensity (24).

Myocardial Fatty Foci

Although the classic cardiac manifestation of tuberous sclerosis is rhabdomyoma, in recent years there have been reports of myocardial fatty deposition, predominantly in older children. There are often multiple fatty deposits, commonly in the ventricular septum, and they can be intramyocardial or epicardial. At echocardiography, myocardial fatty foci appear as multiple circumscribed echogenic mass lesions. At CT, they show low attenuation with no enhancement. Fatty deposits are hyperintense compared with myocardium at T1-weighted and T2-weighted MRI and lose signal intensity during fat-suppressed sequences. At cine SSFP MRI, myocardial fatty foci may demonstrate a rim of chemical shift artifact (24,70).

Differential Diagnosis

The definitive diagnosis of cardiac tumors can only be made at histopathologic evaluation after surgical resection or biopsy; however, some imaging characteristics may suggest a benign cause. There are usually multiple rhabdomyomas, and they appear as homogeneous, hyperechoic, lobulated masses. They are often associated with tuberous sclerosis that does not calcify, and they tend to deform in the opposite direction of the surrounding myocardium when evaluated with echocardiography. Rhabdomyoma is almost always diagnosed prenatally or in early infancy (71) on the basis of clinical and imaging features; thus, biopsy is not required for diagnosis. In comparison, fibromas do not seem to deform or contract in any direction (10), are usually a single mass, and often show central calcification (6,15).

Imaging findings are similar for IMTs and myxomas due to their pedunculated appearance (31). Myxomas appear more commonly in the left atrium with heterogeneous delayed enhancement, whereas the delayed enhancement in an IMT can be persistent and intense. The possibility of a thrombus must also be ruled out given the tendency of sporadic cardiac myxomas to be multicentric and to develop in the left atrium. However, although chronic thrombi may appear to have heterogeneous enhancement at their periphery, thrombi are less likely to have the typical stalklike attachment to the interatrial septum and the pedunculated morphology of myxomas.

Teratomas are heterogeneous masses that can contain fat or calcification on images. A fat-fluid level or calcification in the mass is diagnostic of a mature teratoma. Intracardiac teratomas can be differentiated from other intracardiac tumors due to their multilocular appearance. Fibromas may also contain calcification but tend to be homogeneous in appearance in comparison to the appearance of teratomas. Teratomas can also be extracardiac tumors and can mimic other pericardial tumors if they have a high fat content, including lipomas and

lipoblastomas. However, lipomas and lipoblastomas are typically more homogeneous.

The differential diagnoses for benign vascular masses (hemangiomas) include malignant vascular masses such as angiosarcomas and highly vascularized tumors such as paragangliomas. Although both paragangliomas and hemangiomas are hyperintense on T2-weighted MR images, paragangliomas are hypointense on T1-weighted MR images (72). Unlike hemangiomas, the majority of angiosarcomas occur in the right atrium, and the majority of paragangliomas occur in the roof of the left atrium.

Because of their location, left atrial sarcomas can mimic the appearance of left atrial myxomas. If invasive features are shown, heterogeneous areas of necrosis and hemorrhage are noted, and no stalk is identified, a primary sarcoma should be suspected (9). Primary cardiac lymphomas are usually uniform tumors. Necrosis and involvement of the cardiac valves are atypical and suggest an alternative diagnosis such as angiosarcoma, especially if it manifests in the right atrium (8). Lymphomas are typically homogeneous and centered on the atrioventricular groove, and they spare the right coronary artery. Angiosarcomas are usually centered on the right atrial free wall and have a heterogeneous infiltrative appearance, with areas of necrosis or hemorrhage.

Mediastinal masses such as teratomas and thymolipomas are masses that should be considered when evaluating pericardial lipomas and lipoblastomas. Mediastinal teratomas have fat but also have fluid-attenuation cystic components and calcification, which are features that can help to differentiate these masses from lipomas and lipoblastomas. Thymolipoma, or thymolipomatous hamartoma, is a fatty mass of the thymus that measures an average of 20 cm in diameter at diagnosis. They have soft-tissue nodules and septa and are connected to the thymic bed (73).

Diaphragmatic defects with the herniation of omental fat, such as Morgagni hernias, are also a diagnostic consideration, but multiplanar imaging and the presence of abdominal vasculature should allow differentiation of this diagnosis (62).

Conclusion

Primary cardiac tumors are rare in the pediatric population; however, it is important for radiologists to have an understanding of the most common imaging and epidemiologic characteristics. The most common cardiac mass in children is rhabdomyoma, followed by fibroma. Cardiac tumors in children are frequently associated with specific syndromes, allowing the narrowing of the differential diagnosis. With adequate imaging techniques and knowledge of the pathologic basis, it is often possible to differentiate between benign and malignant neoplasms, which can greatly affect adequate and timely treatment.

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